

The boundary-graviton one-loop determinant as an infinite product of twisted Ruelle zetas, and the three residues that remain

26 August 2026

Abstract

For a hyperbolic 3-manifold $X = \Gamma \backslash \mathbb{H}^3$, the one-loop determinant of boundary gravitons in AdS_3 is conventionally written as a product over primitive conjugacy classes in terms of the nome $q_\gamma = e^{-\ell_\gamma + i\theta_\gamma}$. We observe that the symmetric-power representation σ_k entering the twisted Ruelle zeta at integer argument k is *one-dimensional*, so that

$$\mathcal{R}(k, \sigma_k) = \prod_{[\gamma] \text{ prime}} (1 - q_\gamma^k)$$

with q_γ *exactly* that nome. The boundary-graviton one-loop determinant is therefore

$$Z_{1\text{-loop}} = \prod_{n \geq 2} |\mathcal{R}(n, \sigma_n)|^{-2},$$

an infinite product of twisted Ruelle zeta values. In particular it is *not* any single Ray–Singer analytic torsion: to the question “which of Pfaff’s torsions is it?” the answer is none of them. Pfaff’s ratio formula supplies exactly the tail $k \geq 3$; the one factor it cannot reach is $n = 2$, which sits *at* the abscissa of absolute convergence $\text{Re}(s) > 2$ — which is also why that formula begins at $m \geq 3$ and normalises by $\rho(2)$.

We then state precisely what is not thereby established. Three residues remain, and we separate them rather than merging them into a single caveat: (1) a gauge-fixed spin-2 test function for the finite-volume cusped case, which we believe is a genuine gap in the literature; (2) an evaluation-point gap — Fried’s theorem relates the twisted Ruelle zeta at $s = 0$ to analytic torsion, Pfaff’s ratios live at $k \geq 3$, and the graviton product at $s = n \geq 2$, so these are three evaluations of one family and Fried’s is the one point the graviton never visits; (3) the cusp’s continuous spectrum. For *m004* we also settle a related question negatively: its Ruelle zeta does not factor through $L(s, \chi_{-3})$ in its Euler product, which is purely geodesic, although $L(s, \chi_{-3})$ does enter the functional equation through the cusp. No measured quantity is claimed anywhere.

1 Introduction

1.1 Setting

Let $X = \Gamma \backslash \mathbb{H}^3$ be a hyperbolic 3-manifold. For a primitive closed geodesic $[\gamma]$ of complex length $\ell_\gamma + i\theta_\gamma$, write $q_\gamma = e^{-\ell_\gamma + i\theta_\gamma}$. In the standard treatment of one-loop quantum gravity in AdS_3 , the contribution of boundary gravitons is assembled from products over $[\gamma]$ in powers of q_γ and $\bar{q}_\gamma$.

Separately, for a finite-dimensional representation σ of Γ one has the twisted Ruelle zeta function, defined for $\text{Re}(s)$ large by

$$\mathcal{R}(s, \sigma) = \prod_{[\gamma] \text{ prime}} \det(I - \sigma(\gamma) e^{-s \ell_\gamma}),$$

and continued meromorphically. Ruelle zeta values are related to Ray–Singer analytic torsion by theorems of Fried and, in the setting relevant here, by Pfaff’s ratio formula.

These two objects are usually discussed in different languages. The point of §2 is that in this case they are not merely related — they coincide, term by term, for an elementary reason.

1.2 The observation

The representations σ_k that appear are one-dimensional. When σ is one-dimensional the determinant in the definition of $\mathcal{R}$ is just a scalar, and the $e^{-s\ell_\gamma}$ combines with the character value $e^{ik\theta_\gamma}$ into precisely q_γ^k . There is no room for the two conventions to differ, and the identification is an identity rather than a correspondence. Everything in §2 follows from that observation; we claim no depth for it, only that stating it clearly changes what one can then prove and, more usefully, changes what one can then *locate* as missing.

1.3 What this paper claims, and what it does not

- **Claimed:** the identity (Theorem 2.1); the consequence that the one-loop determinant is an infinite product and not a single torsion (Corollary 2.2); the exact match between Pfaff’s ratio formula and the $k \geq 3$ tail (Proposition 2.3); numerical evidence, on an instrument with a control proved able to detect failure, that the $n = 2$ factor does not break down at the abscissa (§3); and the negative result on $L(s, \chi_{-3})$ (Theorem 5.1).
- **Not claimed:** the assembly of the one-loop partition function. Three specific obstructions remain and are stated as Residues 1–3 in §4, each with what would close it. Convergence of the $n = 2$ factor is *not* proved (Scope 3.2). No numerical value is offered as a measurement of anything.

2 The identity

Theorem 2.1 (the dictionary entry is an identity). *Let σ_k denote the one-dimensional representation sending $\gamma \mapsto e^{ik\theta_\gamma}$. Then for $\operatorname{Re}(s) > 2$,*

$$\mathcal{R}(k, \sigma_k) = \prod_{[\gamma] \text{ prime}} (1 - q_\gamma^k), \quad q_\gamma = e^{-\ell_\gamma + i\theta_\gamma},$$

with q_γ exactly the nome appearing in the boundary-graviton product. Consequently

$$Z_{1\text{-loop}} = \prod_{n \geq 2} |\mathcal{R}(n, \sigma_n)|^{-2}.$$

Proof. Since σ_k is one-dimensional, $\det(I - \sigma_k(\gamma)e^{-k\ell_\gamma}) = 1 - e^{ik\theta_\gamma}e^{-k\ell_\gamma} = 1 - (e^{-\ell_\gamma + i\theta_\gamma})^k = 1 - q_\gamma^k$. The boundary-graviton determinant is the product over $n \geq 2$ of $\prod_\gamma |1 - q_\gamma^n|^{-2}$, which is the displayed expression. $\square$

Corollary 2.2 (it is not a single analytic torsion). *$Z_{1\text{-loop}}$ is an infinite product of Ruelle zeta values. It is therefore not equal to any one Ray–Singer analytic torsion $T_X(\rho)$, nor to any finite ratio of such; asking which torsion it “is” has the answer: none of them.*

This is a negative answer to a natural question, and it is worth stating because the question is asked. The positive content is that the individual factors are torsion-theoretic, which is what Proposition 2.3 makes precise.

Proposition 2.3 (Pfaff’s formula supplies exactly the tail). *With $\rho(m) = \text{Sym}^{2m} \mathbb{C}^2$ and $\kappa(X)$ the number of cusps, Pfaff’s ratio formula reads*

$$\frac{T_X(\rho(m))}{T_X(\rho(2))} = \left(\frac{c(m)}{c(2)} \right)^{\kappa(X)} \exp\left(-\frac{1}{\pi} \text{vol}(X)(m(m+1)-6)\right) \prod_{k=3}^m |\mathcal{R}(k, \sigma_k)|.$$

The product on the right is exactly the $k \geq 3$ portion of Theorem 2.1. The formula therefore expresses the tail of $Z_{1\text{-loop}}$ in torsion-theoretic terms, and does not reach $n = 2$.

Remark 2.4 (why the formula starts at $m \geq 3$). $\mathcal{R}(s, \sigma_k)$ converges absolutely for $\text{Re}(s) > 2$. The factor $n = 2$ therefore sits *at* the abscissa of absolute convergence, not inside it. This is the structural reason the ratio formula begins at $m \geq 3$ and normalises by $\rho(2)$: the excluded factor is exactly the one at the boundary of the region where the Euler product converges absolutely. It is also why naive numerical evaluation of the $n = 2$ term is oscillatory in the geodesic cutoff, while the $n \geq 3$ tail is not.

3 The $n = 2$ factor at the abscissa

Because $n = 2$ is the one factor outside the reach of Proposition 2.3, its status matters. We report a numerical study, and we are explicit that it is numerical.

Proposition 3.1 (no breakdown, on an instrument that can see one). *Approach the abscissa from above, where $\mathcal{R}(s, \sigma_2) = \prod_{\gamma} (1 - e^{2i\theta_{\gamma}} e^{-s\ell_{\gamma}})$ converges absolutely, and track the dependence on the geodesic cutoff. For m004, over three cutoffs and seven values of s , $|\mathcal{R}(s, \sigma_2)|$ is smooth and monotone through $s = 2$ — rising from 1.1075 at $s = 2.6$ to 1.1936 at $s = 2.0$ — with no pole and no discontinuity, and with a cutoff spread that grows smoothly rather than diverging.*

Scope 3.2 (what this does and does not establish). The reading is meaningful only because the instrument is shown able to report failure: taken *below* the abscissa, where divergence is certain, the same computation does diverge and the cutoff spread blows up. That is the bite control, and without it a smooth curve would be evidence of nothing.

What is established is the absence of any observable breakdown at $s = 2$. What is *not* established is convergence. Three cutoffs are used and the cutoff-to-infinity limit is untested; a spread that is small at cutoff 5.5 may still fail to converge. Accordingly the status of the $n = 2$ factor moves from “may not exist” to “no evidence of breakdown; existence unproved”, and it remains one of the three residues below.

Remark 3.3 (a consistency check on the assembly). Computing $\log Z$ for the assembled $n \geq 3$ tail in two independent ways — once through the Ruelle factors $\mathcal{R}(k, \sigma_k)$ and once directly through the geodesic sum over $[\gamma]$ — agrees to 8.2×10^{-14} . This tests the identity of Theorem 2.1 in the regime where both sides converge absolutely, rather than testing arithmetic.

4 The three residues

A single caveat covering everything unfinished is not useful to a reader. We separate the obstructions, because they have different characters and different remedies, and because two of the three are not what one would first guess.

Residue 1 (the spin-2 cusped test function). Assembling the one-loop determinant for a finite-volume cusped hyperbolic 3-manifold requires a gauge-fixed spin-2 (and vector, and scalar) one-loop determinant in the cusped setting. We have not found this in the literature, and we believe it to be a genuine gap rather than a retrieval failure: adjacent literature

returns richly for the compact case and for the $\mathrm{SL}(2, \mathbb{C})$ and $\mathrm{SL}(n, \mathbb{C})$ state-integral families, while this specific combination does not appear. **What would close it:** a construction and evaluation of that determinant, or a citation we have missed.

Residue 2 (the evaluation point). Fried's theorem gives, for acyclic orthogonal representations, that the twisted Ruelle zeta is holomorphic at 0 with $\mathcal{R}_\sigma(0) = T_X(\sigma)^2$, and a cusped extension exists in the literature via a regularised trace. It is natural to expect this closes the gap between the graviton determinant and analytic torsion. **It does not, and the reason is the evaluation point.** Fried's identity lives at $s = 0$; Pfaff's ratios live at $k \geq 3$; the graviton product lives at $s = n \geq 2$. These are three evaluations of one twisted Ruelle family, and Fried's is the single point the graviton product never visits.

The gap can nevertheless be located exactly. Since $\rho(m) = \mathrm{Sym}^{2m} \mathbb{C}^2$ has eigenvalues e^{jL} for $j = -m, \dots, m$ on a holonomy of complex length $L = \ell + i\theta$, and $e^{jL}e^{-s\ell} = e^{ij\theta}e^{-(s-j)\ell}$, one has the identity

$$\mathcal{R}_{\rho(m)}(s) = \prod_{j=-m}^m \mathcal{R}(s-j, \sigma_j), \quad (1)$$

which is elementary and we claim no novelty for it. At $s = 0$ it reads

$$\mathcal{R}_{\rho(m)}(0) = \mathcal{R}(0, \sigma_0) \cdot \underbrace{\prod_{j=1}^m \overline{\mathcal{R}(j, \sigma_j)}}_{\text{the graviton's own factors}} \cdot \underbrace{\prod_{j=1}^m \mathcal{R}(-j, \sigma_j)}_{\text{negative arguments}}.$$

The graviton's factors sit inside Fried's point explicitly; the obstruction is the second product, at arguments outside absolute convergence. **Residue 2 therefore reduces to evaluating $\mathcal{R}$ at the negative integers** — and since a reflection $s \leftrightarrow 2-s$ carries $2+j$ to $-j$, with $\mathcal{R}(2+j, \sigma_j)$ absolutely convergent for $j \geq 1$, it reduces again:

The gap can nevertheless be located exactly, and one route to closing it can be ruled out. Taking the ratio of (1) at m and $m-1$ isolates $\mathcal{R}(-m, \sigma_m) \cdot \overline{\mathcal{R}(m, \sigma_m)}$. If one could write $\mathcal{R}_{\rho(m)}(0) = T_X(\rho(m))^2$ — Fried's theorem — then Proposition 2.3 would evaluate that ratio and give a reflection formula expressing $|\mathcal{R}(-m, \sigma_m)|$ through the absolutely convergent $|\mathcal{R}(m, \sigma_m)|$.

That route is closed. Fried's hypothesis is acyclicity, and here it fails at every m :

Proposition 4.1 (the coefficients are never acyclic). *For the figure-eight knot complement and $\rho(m) = \mathrm{Sym}^{2m} \mathbb{C}^2$,*

$$(h^0, h^1, h^2) = (0, 1, 1) \quad \text{for every } m \geq 1, \quad (1, 1, 0) \quad \text{for } m = 0.$$

The obstruction is peripheral and independent of m : the cusp's parabolic subgroup fixes exactly one line in $\mathrm{Sym}^n \mathbb{C}^2$ for every n , which forces $h^1 = h^2 = \kappa(X) = 1$.

Consequently $\mathcal{R}_{\rho(m)}(s)$ has a nonzero order of vanishing at $s = 0$ rather than a finite value equal to a torsion, and the reflection formula does *not* follow by this route.

What would close it: the cusped Ray–Singer theory of Park and Pfaff, whose acyclicity hurdle is met by the canonical one-dimensional cusp-cohomology basis, with the *leading Laurent coefficient* at $s = 0$ in place of the value. **We have not carried out that derivation, and residue 2 remains open.** What is unconditional is (1) alone.

Residue 3 (the cusp's continuous spectrum). The continuous spectrum contributed by the cusp is not included in the assembly above. Its scattering determinant for $m004$ is available in closed form, but incorporating it into the one-loop object is not done here. **What would close it:** the assembly, once Residue 1 is available.

Remark 4.2 (a correction we record). We initially expected Fried’s theorem to close Residue 2 outright, and said so before checking where each theorem evaluates. It does not. We record the expectation and its failure because the reframing in Residue 2 is the useful output, and it was only reached by asking which point of the family each theorem actually speaks about.

5 What the Ruelle zeta does not know about $L(s, \chi_{-3})$

The manifold $m004$ has invariant trace field $\mathbb{Q}(\sqrt{-3})$, and the Dirichlet L -function $L(s, \chi_{-3})$ appears in its volume. It is natural to ask whether the Ruelle zeta factors through that L -function. It does not.

Theorem 5.1. *The Euler product of $\mathcal{R}(k, \sigma_k)$ is purely geodesic: by Theorem 2.1 it is $\prod_{\gamma}(1 - q_{\gamma}^k)$, indexed by primitive closed geodesics, and it contains no L -function factor. In particular $\mathcal{R}$ does not factor through $L(s, \chi_{-3})$ in its Euler product. However, $L(s, \chi_{-3})$ does enter the functional equation, through the scattering determinant of the cusp.*

The distinction is worth preserving. Arithmetic enters this object through the cusp and the functional equation, not through the geodesic Euler product; a claim that the Ruelle zeta “is” an L -function, or a ratio of them, is false in the Euler-product sense in which such claims are usually meant.

6 Independent corroboration

The two central negative findings of this paper — that the cited torsion formulae and the scalar cusp scattering determinant do not by themselves supply the one-loop object, and that a gauge-fixed spin-2 determinant for the finite-volume cusped case is not available — were reached independently, by different means, and recorded in a register of questions as **REFUTED** and **EXTERNAL_BLOCKER** respectively (Appendix B).

We note this because the second is an assertion of absence, and assertions of absence are the weakest kind of claim a single investigator can make. Two independent arrivals at the same boundary — agreeing not only that something is missing but on *which* thing is missing and which adjacent things are present — is materially better evidence than one, and it is the reason we are willing to state Residue 1 as a gap rather than as a failure to find.

7 What is not claimed

No measured quantity appears in this paper, and no identification of any quantity here with any observable is offered or used. The numerical work of §3 is evidence about convergence behaviour, not a computation of a physical value. Convergence of the $n = 2$ factor is unproved (Scope 3.2), and the one-loop partition function is not assembled (§4).

A Verification

The computations are reproduced by the scripts in **verify/**, over the complex length spectrum of $m004$. Controls run before results are read.

discharges	control or check
Thm 2.1	$\log Z$ for the $n \geq 3$ tail computed twice, once via $\mathcal{R}(k, \sigma_k)$ and once via the direct geodesic sum, agreeing to 8.2×10^{-14} ; conjugate pairs identified by distance to the nearest multiple of 2π , not by a raw modulus, and the four self-conjugate $\theta = \pi$ classes handled separately
Prop. 2.3	the torsion ratios at $m = 3, 4, 5, 6$ reproduce independently computed values; $\kappa(X) = 1$; $\text{vol}(X) = 2.029883212819307$ taken at full double precision, since a truncated decimal string manufactures a spurious mismatch at the 10^{-11} level
Prop. 3.1	bite control: the same evaluation taken <i>below</i> the abscissa must diverge with a blowing-up cutoff spread, and does; the reported smoothness above the abscissa is meaningful only against it
Thm 5.1	the Euler product is regenerated from the geodesic list rather than quoted, and checked against a Dirichlet-series evaluation performed by analytic continuation — a naive series at $s = 1/2$ lies outside its half-plane and produces a spurious refutation

B Status in the programme register

entry	status	relation to this paper
OA-C1061	REFUTED	whether the cited Fried, Park or Pfaff torsion formulae, or the scalar cusp scattering determinant, directly supply the one-loop object. Independently reached; matches Corollary 2.2 and Residue 2.
OA-C1062	EXTERNAL_BLOCKER	whether a gauge-fixed spin-2/vector/scalar one-loop determinant can be constructed for the finite-volume cusped case. Independently reached; this is Residue 1.
OA-C1059	REFUTED	whether the untwisted Ruelle zeta is a finite product or ratio of shifted Dirichlet L -functions. Consistent with Theorem 5.1.
OA-C1060	EXTERNAL_BLOCKER	whether the $n = 2$ factor has a proved cutoff-independent value. This is exactly Scope 3.2: we report no breakdown and do not claim existence.

Three of the four are negative or blocked. We regard that as the honest shape of this subject at present, and prefer to publish the identity together with an accurate map of what it does not close.

References

- [1] D. Fried, Analytic torsion and closed geodesics on hyperbolic manifolds, *Invent. Math.* **84** (1986), 523–540.
- [2] J. Pfaff, Analytic torsion versus Reidemeister torsion on hyperbolic 3-manifolds with cusps, arXiv:1206.0228.
- [3] J. Park, Analytic torsion and Ruelle zeta functions for hyperbolic manifolds with cusps, *J. Funct. Anal.* **257** (2009), 1713–1758.
- [4] S. Giombi, A. Maloney and X. Yin, One-loop partition functions of 3d gravity, *JHEP* **08** (2008), 007.
- [5] A. Maloney and E. Witten, Quantum gravity partition functions in three dimensions, *JHEP* **02** (2010), 029.
- [6] R. B. Melrose, *The Atiyah–Patodi–Singer Index Theorem*, A K Peters, 1993.